ZIP Analysis: Zero-Inflation and Its Misspecification

STAT778 Project

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Contents

Purpose

Analyzes the ZIP (zero-inflated Poisson) study. Three generating levels of zero-inflation (p0 = 0, 0.2, 0.4), each fit two ways:

- **Well-specified:** fit as ZIP.
- **Misspecified:** fit as Poisson (ignoring the excess zeros).

Question: does ignoring zero-inflation systematically bias estimates? Does GHK cope better than GQT, or vice versa?

Load and tidy

```
source("code/R/logging.R")
source("code/R/io.R")
source("code/R/fit_core.R")

DATASET <- "ZIP"
inventory <- list_fit_results(DATASET, dir = "data/fits")
inventory
```

```
##   scenario_id misspec_family
## 1        p0_0        poisson
## 2        p0_0           <NA>
## 3       p0_20        poisson
## 4       p0_20           <NA>
## 5       p0_40        poisson
## 6       p0_40           <NA>
```

The inventory should show 6 rows: 3 scenarios × {ZIP fit, Poisson fit}. The `misspec_family` column is NA for ZIP fits and `"poisson"` for misspecified fits.

```
fits <- lapply(seq_len(nrow(inventory)), function(i) {
  load_fit_result(
    DATASET,
```

```r
    inventory$scenario_id[i],
    misspec_family = if (is.na(inventory$misspec_family[i])) NULL else inventory$misspec_family[i]
  )
})
names(fits) <- paste(inventory$scenario_id,
                     ifelse(is.na(inventory$misspec_family),
                            "wellspec", inventory$misspec_family),
                     sep = "__")
names(fits)
```

```
## [1] "p0_0__poisson"   "p0_0__wellspec"  "p0_20__poisson"  "p0_20__wellspec"
## [5] "p0_40__poisson"  "p0_40__wellspec"
```

Tidy to long

```r
tidy_one <- function(f) {
  r   <- f$results
  cfg <- f$scenario_config

  rows <- list()
  for (i in seq_len(nrow(r))) {
    if (!r$converged[i]) next
    est <- r$estimates[[i]]
    if (is.null(est) || length(est) == 0) next
    for (pname in names(est)) {
      rows[[length(rows) + 1]] <- data.frame(
        scenario_id   = f$scenario_id,
        p0            = cfg$marginal$od,
        fit_family    = f$fit_marginal_spec$family,
        misspecified  = f$misspecified,
        replicate     = r$replicate[i],
        method        = r$method[i],
        runtime_sec   = r$runtime_sec[i],
        parameter     = pname,
        estimate      = as.numeric(est[[pname]]),
        stringsAsFactors = FALSE
      )
    }
  }
  do.call(rbind, rows)
}

long <- do.call(rbind, lapply(fits, tidy_one))
nrow(long)
```

```
## [1] 5994
```

```r
head(long)
```

```
##                   scenario_id p0 fit_family misspecified replicate method
## p0_0__poisson.1          p0_0  0    poisson         TRUE         1    GHK
## p0_0__poisson.2          p0_0  0    poisson         TRUE         1    GHK
## p0_0__poisson.3          p0_0  0    poisson         TRUE         1    GQT
## p0_0__poisson.4          p0_0  0    poisson         TRUE         1    GQT
## p0_0__poisson.5          p0_0  0    poisson         TRUE         2    GHK
```

```
## p0_0__poisson.6        p0_0  0    poisson         TRUE         2    GHK
##                   runtime_sec parameter  estimate
## p0_0__poisson.1  0.72013807 Intercept 1.1375038
## p0_0__poisson.2  0.72013807     range 0.2295566
## p0_0__poisson.3  0.09797287 Intercept 1.1542940
## p0_0__poisson.4  0.09797287     range 0.2329889
## p0_0__poisson.5  0.53413057 Intercept 1.7236311
## p0_0__poisson.6  0.53413057     range 0.3312742
```

Attach truth

```
long$truth <- NA_real_
long$truth[long$parameter %in% "Intercept"] <- log(5)   # lambda = 5
long$truth[long$parameter %in% "range"]     <- 0.3

long$abs_err <- abs(long$estimate - long$truth)
long$err     <- long$estimate - long$truth
```

Convergence by cell

```
conv <- aggregate(
  converged ~ p0 + fit_family + method,
  data = unique(do.call(rbind, lapply(fits, function(f) {
    cbind(p0 = f$scenario_config$marginal$od,
          fit_family = f$fit_marginal_spec$family,
          f$results[, c("replicate", "method", "converged")])
  }))),
  FUN = function(x) mean(x)
)
conv
```

```
##     p0 fit_family method converged
## 1  0.0    poisson    GHK     1.000
## 2  0.2    poisson    GHK     1.000
## 3  0.4    poisson    GHK     1.000
## 4  0.0        zip    GHK     1.000
## 5  0.2        zip    GHK     1.000
## 6  0.4        zip    GHK     1.000
## 7  0.0    poisson    GQT     1.000
## 8  0.2    poisson    GQT     1.000
## 9  0.4    poisson    GQT     1.000
## 10 0.0        zip    GQT     1.000
## 11 0.2        zip    GQT     0.995
## 12 0.4        zip    GQT     0.995
```

MAE comparison

```
mae_tab <- aggregate(
  abs_err ~ p0 + fit_family + method + parameter,
  data = long,
```

```
  FUN   = mean
)
names(mae_tab)[5] <- "MAE"

mae_tab[order(mae_tab$parameter, mae_tab$p0, mae_tab$fit_family,
              mae_tab$method), ]
```

```
##     p0 fit_family method parameter        MAE
## 1  0.0    poisson    GHK Intercept 0.15346563
## 7  0.0    poisson    GQT Intercept 0.15507243
## 4  0.0        zip    GHK Intercept 0.15381148
## 10 0.0        zip    GQT Intercept 0.15504983
## 2  0.2    poisson    GHK Intercept 0.22057209
## 8  0.2    poisson    GQT Intercept 0.23532575
## 5  0.2        zip    GHK Intercept 0.21840104
## 11 0.2        zip    GQT Intercept 0.20307351
## 3  0.4    poisson    GHK Intercept 0.28733302
## 9  0.4    poisson    GQT Intercept 0.30447581
## 6  0.4        zip    GHK Intercept 0.28460822
## 12 0.4        zip    GQT Intercept 0.22849254
## 13 0.0    poisson    GHK     range 0.07493168
## 19 0.0    poisson    GQT     range 0.07328651
## 16 0.0        zip    GHK     range 0.07785250
## 22 0.0        zip    GQT     range 0.07639521
## 14 0.2    poisson    GHK     range 0.19242271
## 20 0.2    poisson    GQT     range 0.19291717
## 17 0.2        zip    GHK     range 0.10454158
## 23 0.2        zip    GQT     range 0.12478881
## 15 0.4    poisson    GHK     range 0.21022869
## 21 0.4    poisson    GQT     range 0.20978877
## 18 0.4        zip    GHK     range 0.14406065
## 24 0.4        zip    GQT     range 0.30379178
```

Wide view: well-specified vs misspecified, side by side

```
mae_wide <- reshape(
  mae_tab,
  idvar = c("p0", "method", "parameter"),
  timevar = "fit_family",
  direction = "wide"
)
mae_wide$diff <- mae_wide$MAE.poisson - mae_wide$MAE.zip
mae_wide$pct  <- 100 * mae_wide$diff / mae_wide$MAE.zip
mae_wide[order(mae_wide$parameter, mae_wide$p0, mae_wide$method), ]
```

```
##     p0 method parameter MAE.poisson    MAE.zip          diff         pct
## 1  0.0    GHK Intercept  0.15346563 0.15381148 -0.0003458499  -0.22485314
## 7  0.0    GQT Intercept  0.15507243 0.15504983  0.0000225995   0.01457564
## 2  0.2    GHK Intercept  0.22057209 0.21840104  0.0021710527   0.99406700
## 8  0.2    GQT Intercept  0.23532575 0.20307351  0.0322522411  15.88205226
## 3  0.4    GHK Intercept  0.28733302 0.28460822  0.0027248041   0.95738771
## 9  0.4    GQT Intercept  0.30447581 0.22849254  0.0759832733  33.25415950
## 13 0.0    GHK     range  0.07493168 0.07785250 -0.0029208179  -3.75173286
## 19 0.0    GQT     range  0.07328651 0.07639521 -0.0031087006  -4.06923503
```

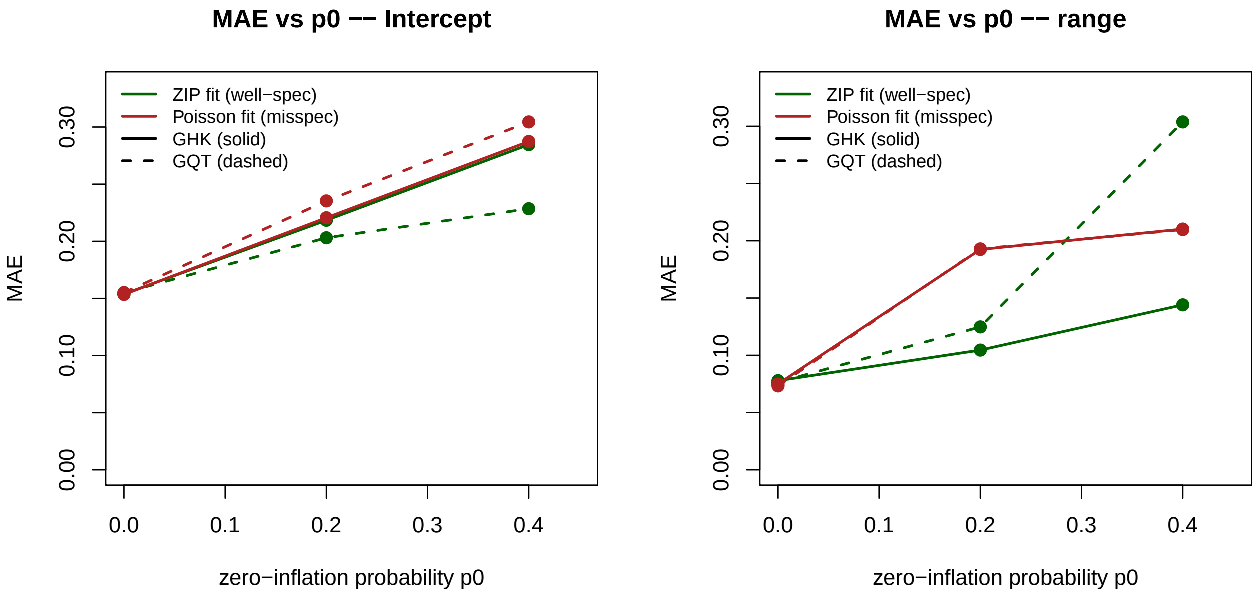
```
## 14 0.2    GHK     range  0.19242271 0.10454158  0.0878811284  84.06332135
## 20 0.2    GQT     range  0.19291717 0.12478881  0.0681283602  54.59492641
## 15 0.4    GHK     range  0.21022869 0.14406065  0.0661680448  45.93068658
## 21 0.4    GQT     range  0.20978877 0.30379178 -0.0940030059 -30.94323573
```

Report claim shape: "Fitting Poisson to ZIP data at p0 = 0.4 inflates MAE on the intercept by roughly X%…"

Figure

```r
par(mfrow = c(1, 2))

for (param in c("Intercept", "range")) {
  d <- mae_tab[mae_tab$parameter == param, ]
  y_max <- max(d$MAE) * 1.1
  plot(NA, xlim = c(0, 0.45), ylim = c(0, y_max),
       xlab = "zero-inflation probability p0",
       ylab = "MAE",
       main = sprintf("MAE vs p0 -- %s", param))
  for (ff in c("zip", "poisson")) {
    for (m in c("GHK", "GQT")) {
      sub <- d[d$fit_family == ff & d$method == m, ]
      sub <- sub[order(sub$p0), ]
      col  <- if (ff == "zip") "darkgreen" else "firebrick"
      lty  <- if (m == "GHK") 1 else 2
      lines(sub$p0, sub$MAE, col = col, lty = lty, lwd = 2)
      points(sub$p0, sub$MAE, col = col, pch = 19, cex = 1.2)
    }
  }
  legend("topleft",
         legend = c("ZIP fit (well-spec)", "Poisson fit (misspec)",
                    "GHK (solid)", "GQT (dashed)"),
         col = c("darkgreen", "firebrick", "black", "black"),
         lty = c(1, 1, 1, 2), lwd = 2, bty = "n", cex = 0.85)
}
```



```r
par(mfrow = c(1, 1))
```

Save

```r
dir.create("visualization/zip", recursive = TRUE, showWarnings = FALSE)
pdf("visualization/zip/mae_vs_p0.pdf", width = 10, height = 5)
# <paste the plot_mae code here>
dev.off()
```

Summary

```r
cat("=== ZIP study summary ===\n\n")
```

```
## === ZIP study summary ===
```

```r
for (p0_val in c(0, 0.2, 0.4)) {
  cat(sprintf("\n-- p0 = %g --\n", p0_val))
  for (param in c("Intercept", "range")) {
    zip_mae <- mean(long$abs_err[long$p0 == p0_val &
                                 long$fit_family == "zip" &
                                 long$parameter == param],
                    na.rm = TRUE)
    pois_mae <- mean(long$abs_err[long$p0 == p0_val &
                                  long$fit_family == "poisson" &
                                  long$parameter == param],
                     na.rm = TRUE)
    cat(sprintf("  %-9s  ZIP fit MAE=%.4f  Poisson fit MAE=%.4f  (ratio=%.2fx)\n",
                param, zip_mae, pois_mae, pois_mae / zip_mae))
  }
}
```

```
##
## -- p0 = 0 --
##   Intercept  ZIP fit MAE=0.1544  Poisson fit MAE=0.1543  (ratio=1.00x)
##   range      ZIP fit MAE=0.0771  Poisson fit MAE=0.0741  (ratio=0.96x)
##
## -- p0 = 0.2 --
##   Intercept  ZIP fit MAE=0.2108  Poisson fit MAE=0.2279  (ratio=1.08x)
##   range      ZIP fit MAE=0.1146  Poisson fit MAE=0.1927  (ratio=1.68x)
##
## -- p0 = 0.4 --
##   Intercept  ZIP fit MAE=0.2566  Poisson fit MAE=0.2959  (ratio=1.15x)
##   range      ZIP fit MAE=0.2237  Poisson fit MAE=0.2100  (ratio=0.94x)
```

Likely findings for the report

Fill in:

1. At p0 = 0 (no zero-inflation), ZIP and Poisson fits produce equivalent MAE (both well-specified).
2. At p0 > 0, Poisson fits _____ the Intercept by _____% more MAE than the ZIP fit.
3. Range estimates are _____ (possibly less affected than Intercept, since zero-inflation is a marginal property).
4. GHK and GQT degrade _____ (together / differently).